

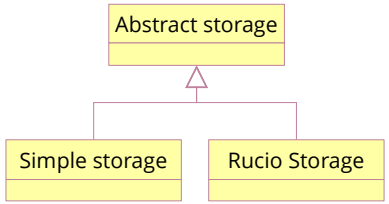
uk | SRC

SKAO Regional Centre United Kingdom

ukSRC Execution Broker

Progress report

Dave
Morris
Manchester
University



OPENAPI

spring®

Java

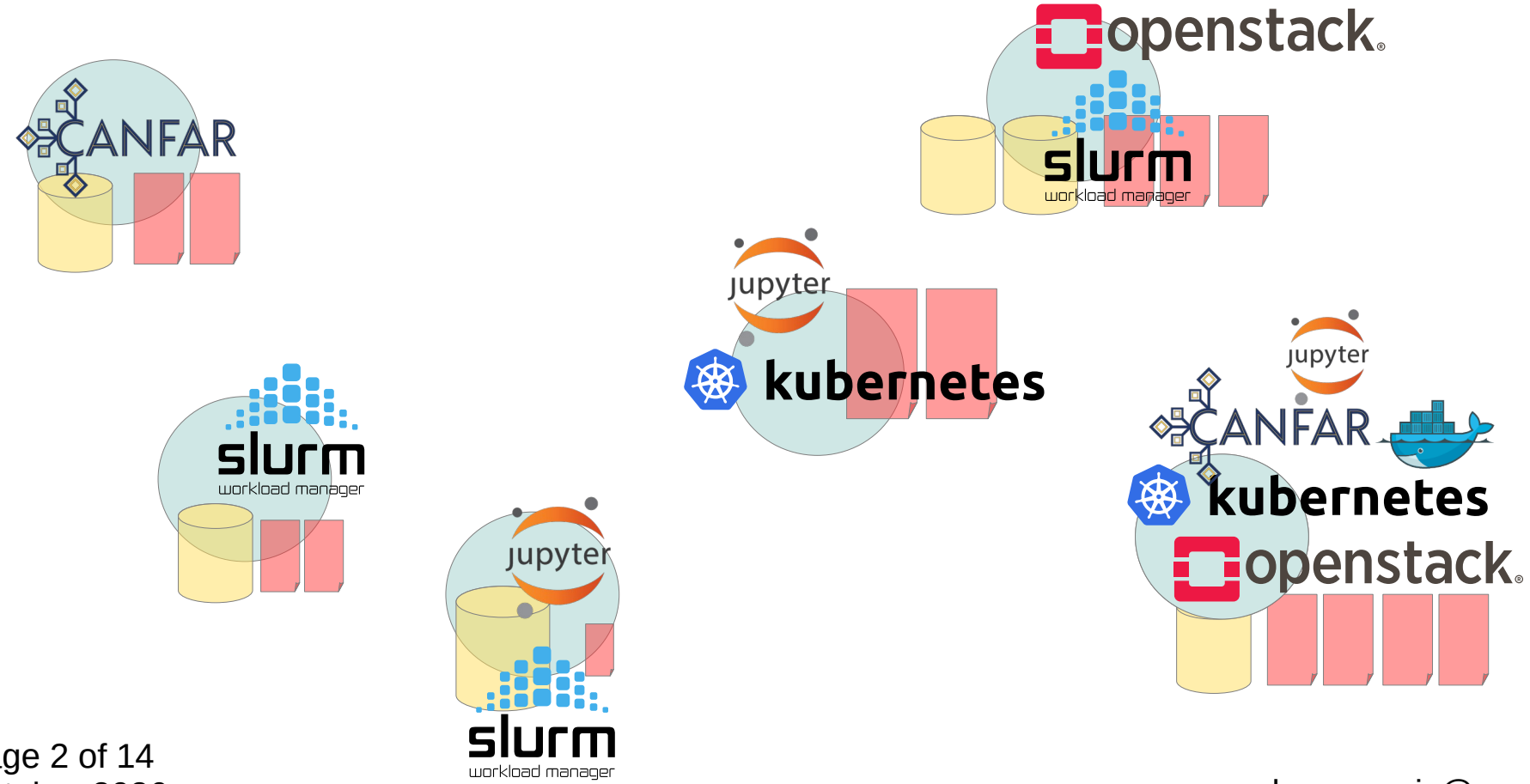
October 2026

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The problem

Lots of different execution platforms

Each with their own local capabilities and policies

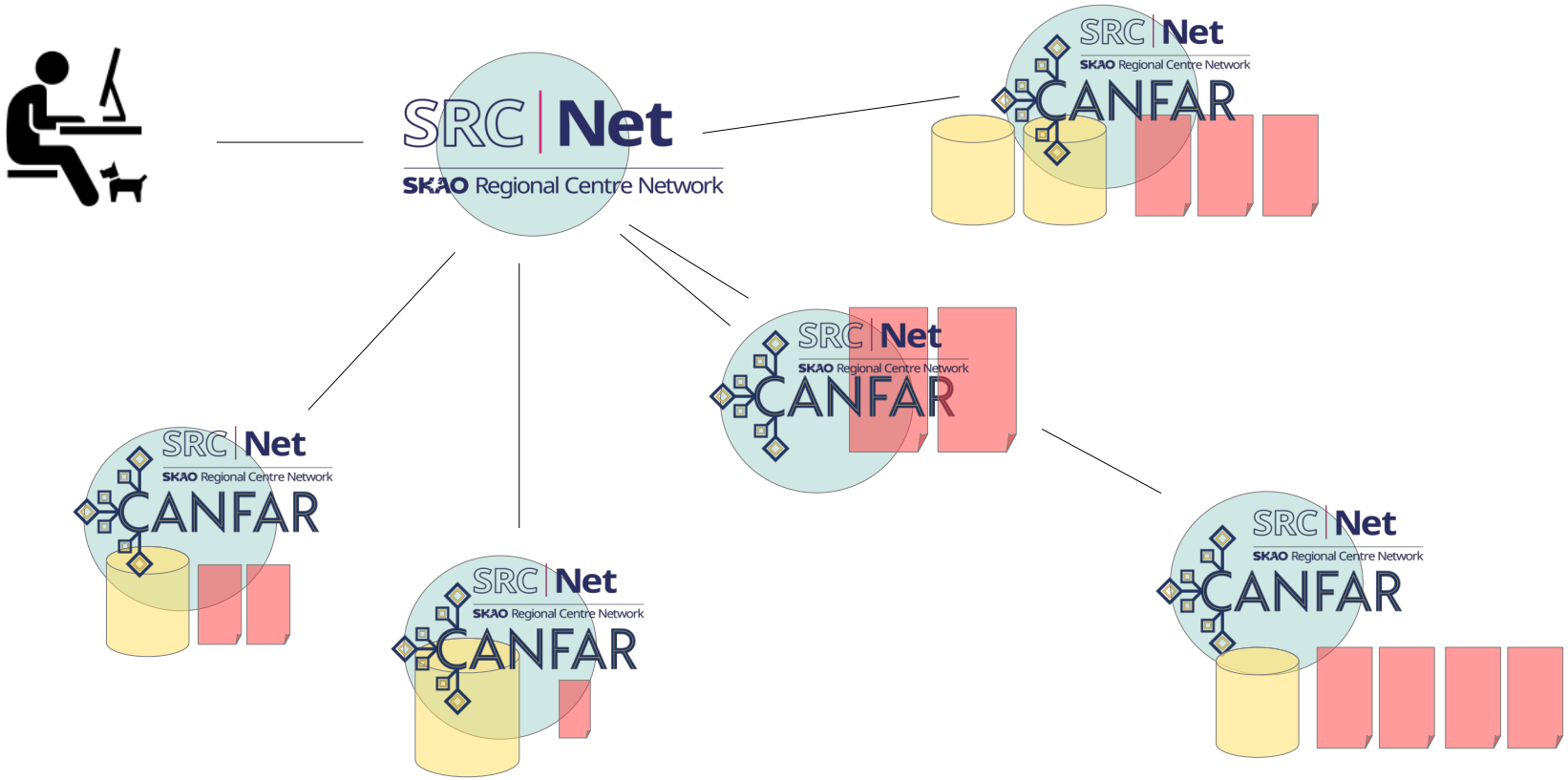


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A solution - homogeneous platforms

All the platforms provide the same services

Centralized command and control

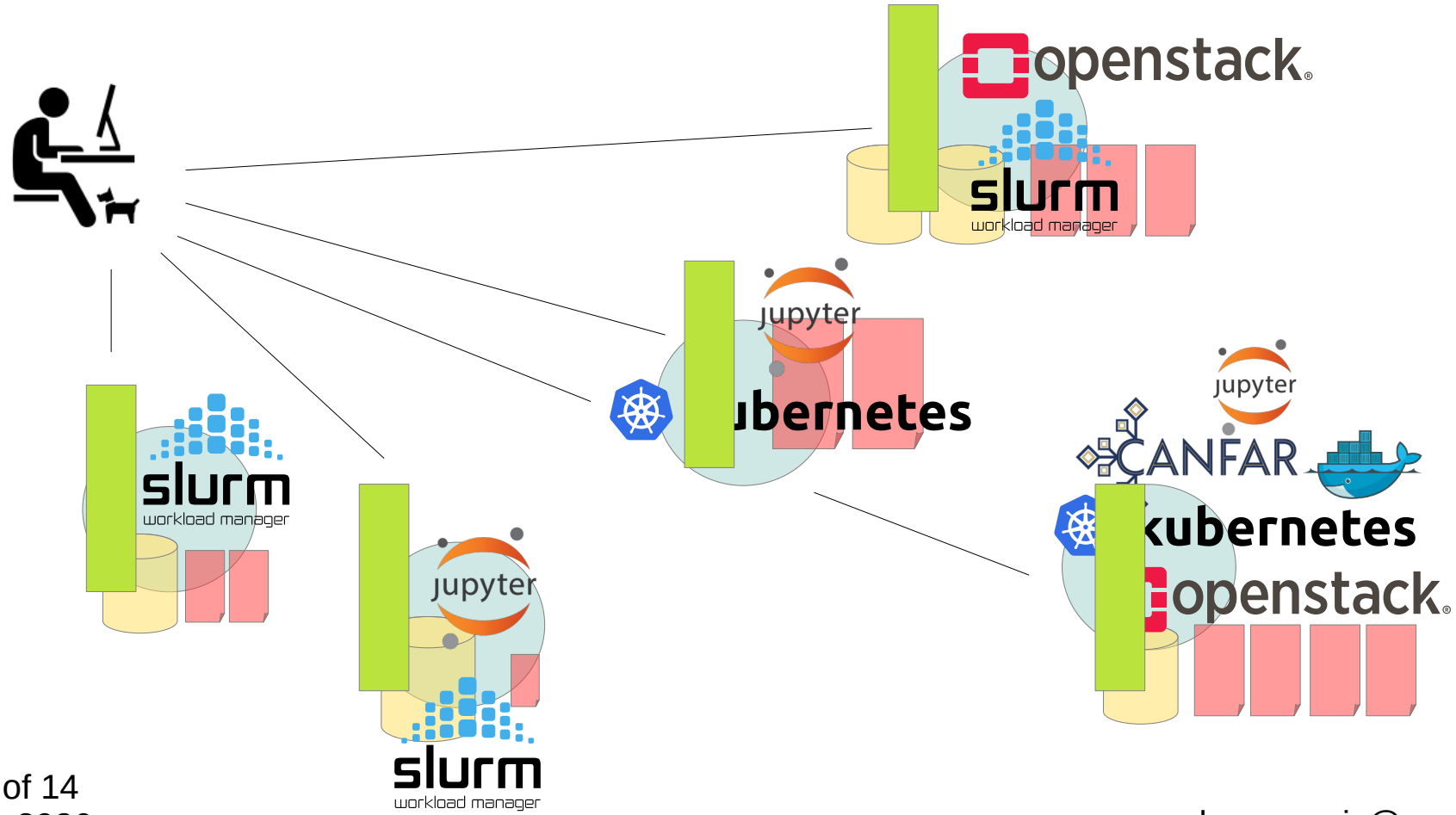


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A solution - homogeneous interface

All the platforms provide the same interface

Distributed control, local policies

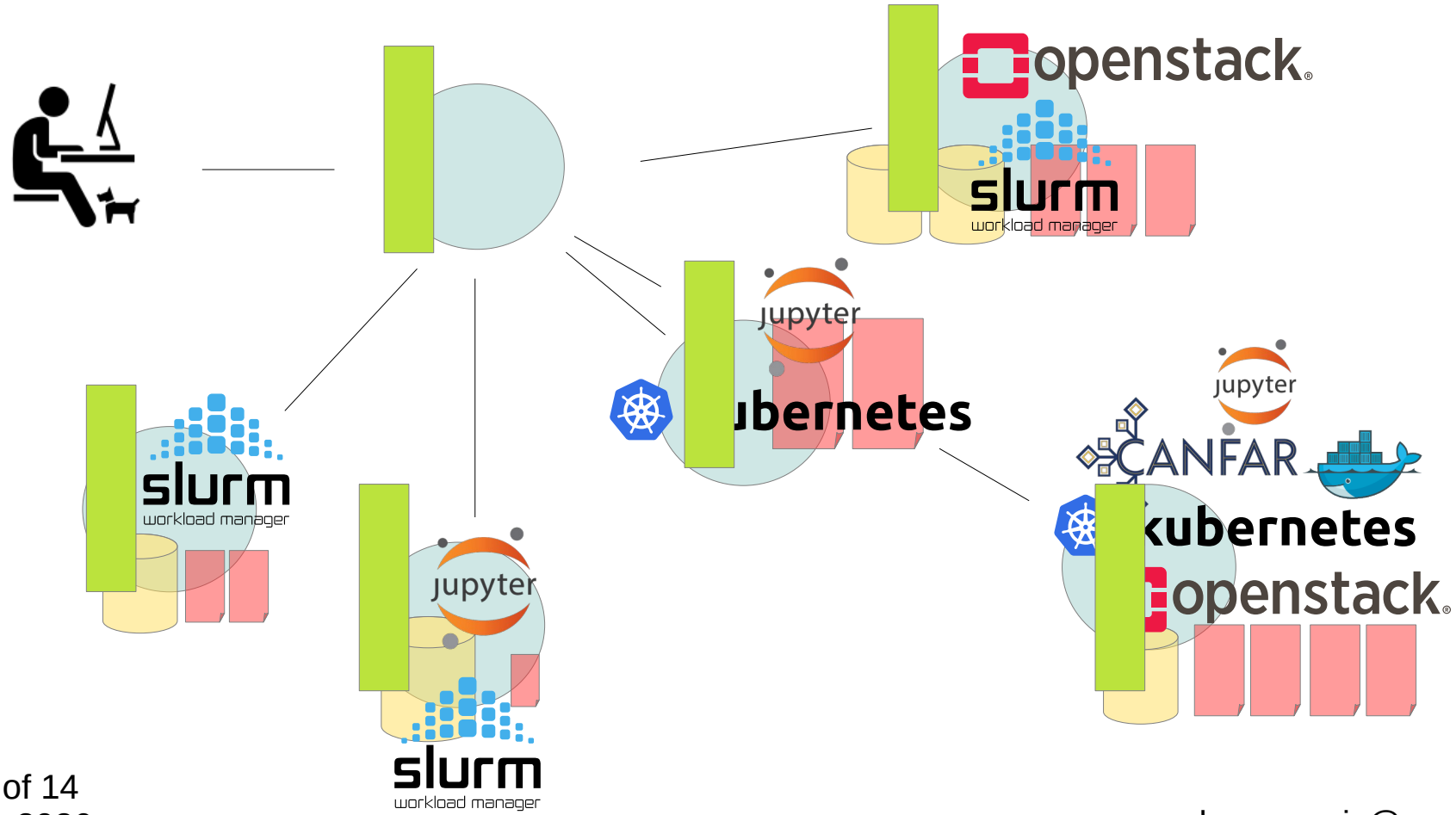


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A solution - homogeneous interface

All the platforms provide the same interface

Distributed control, local policies, aggregator service

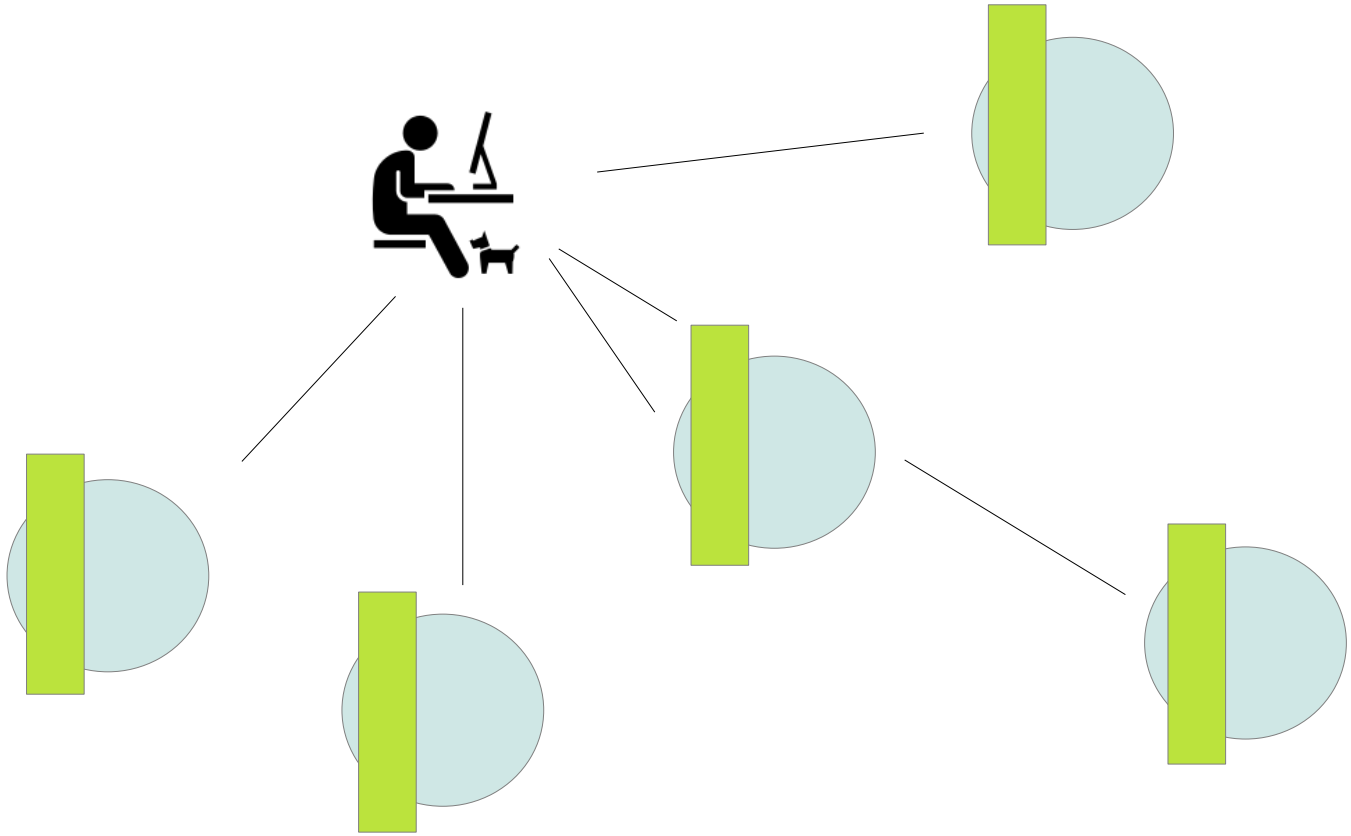


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Implementation

OpenAPI interface

Auto generated Python and Java client libraries

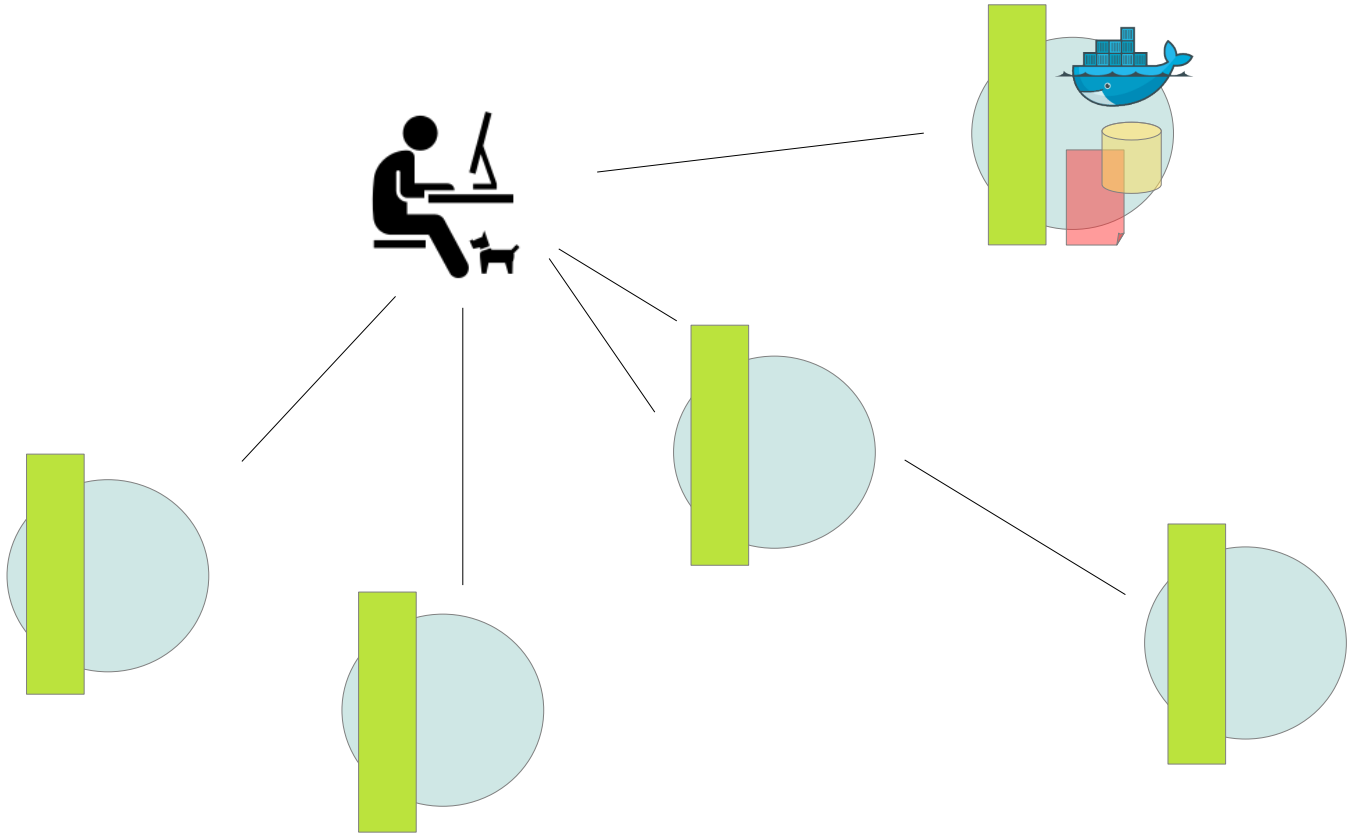


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Implementation

OpenAPI interface

Compute resources with Docker plugin



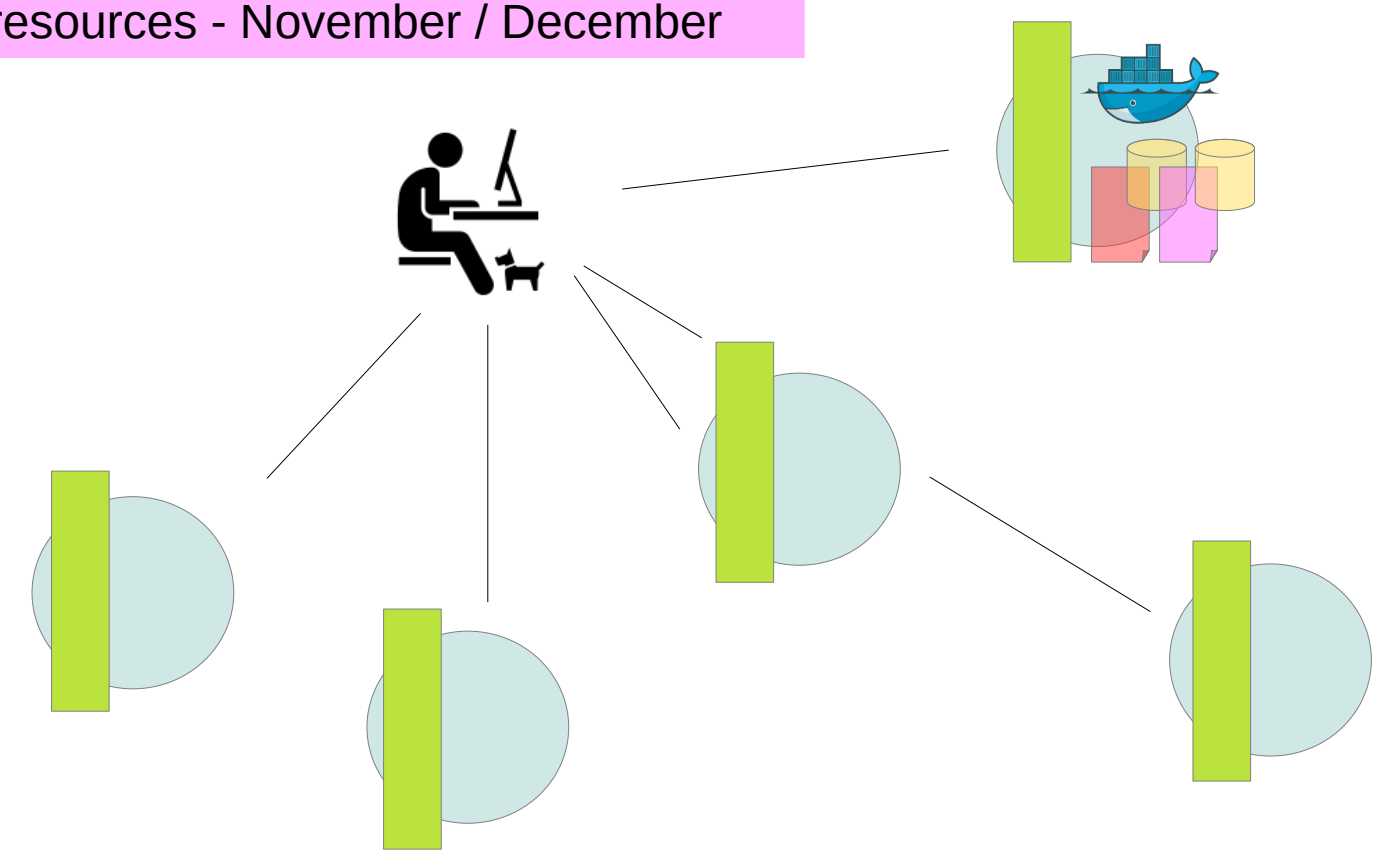
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Implementation

OpenAPI interface

Compute resources with Docker plugin

Storage resources - November / December



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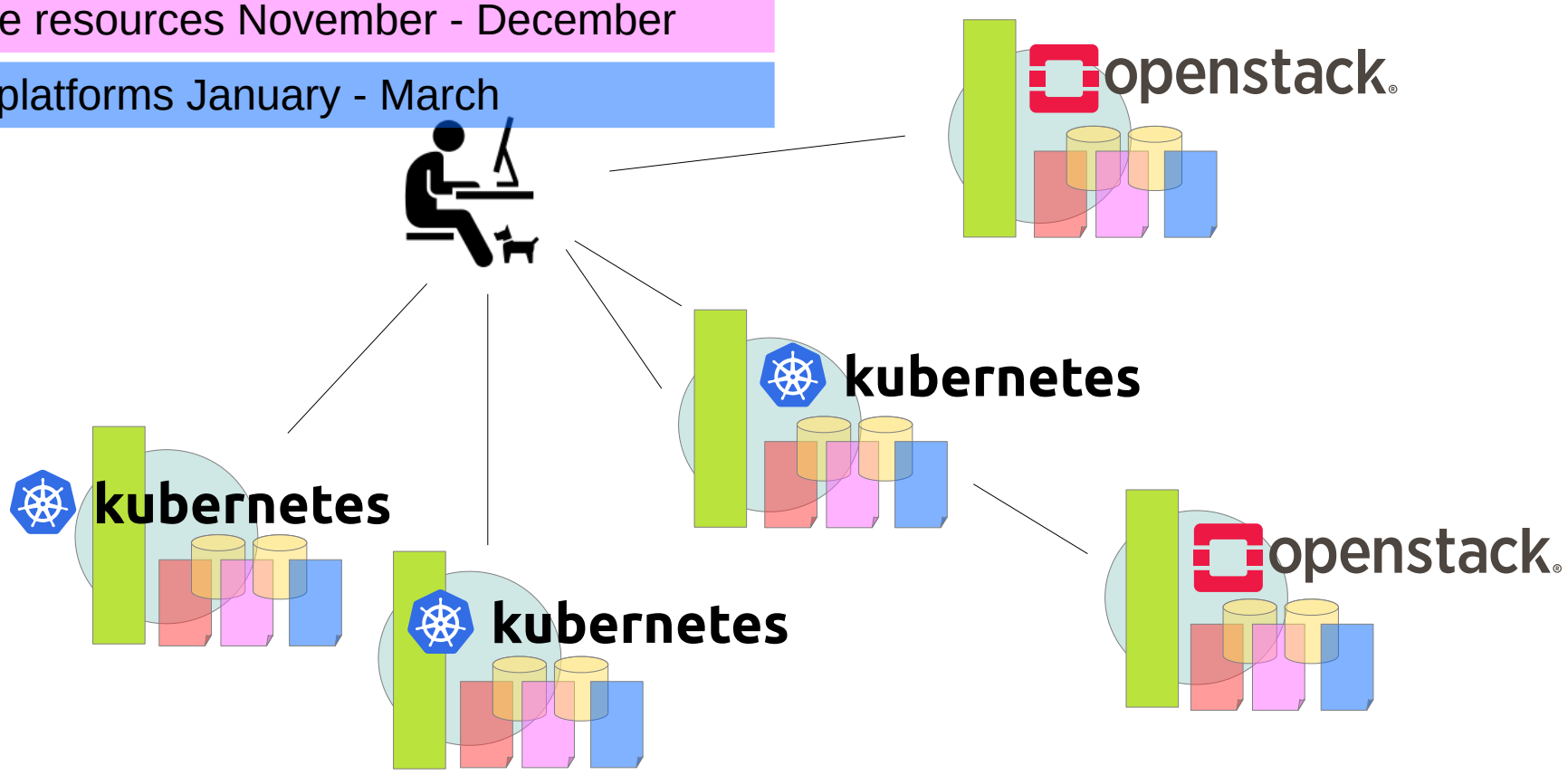
Implementation

OpenAPI interface

Compute resources with Docker plugin

Storage resources November - December

Cloud platforms January - March



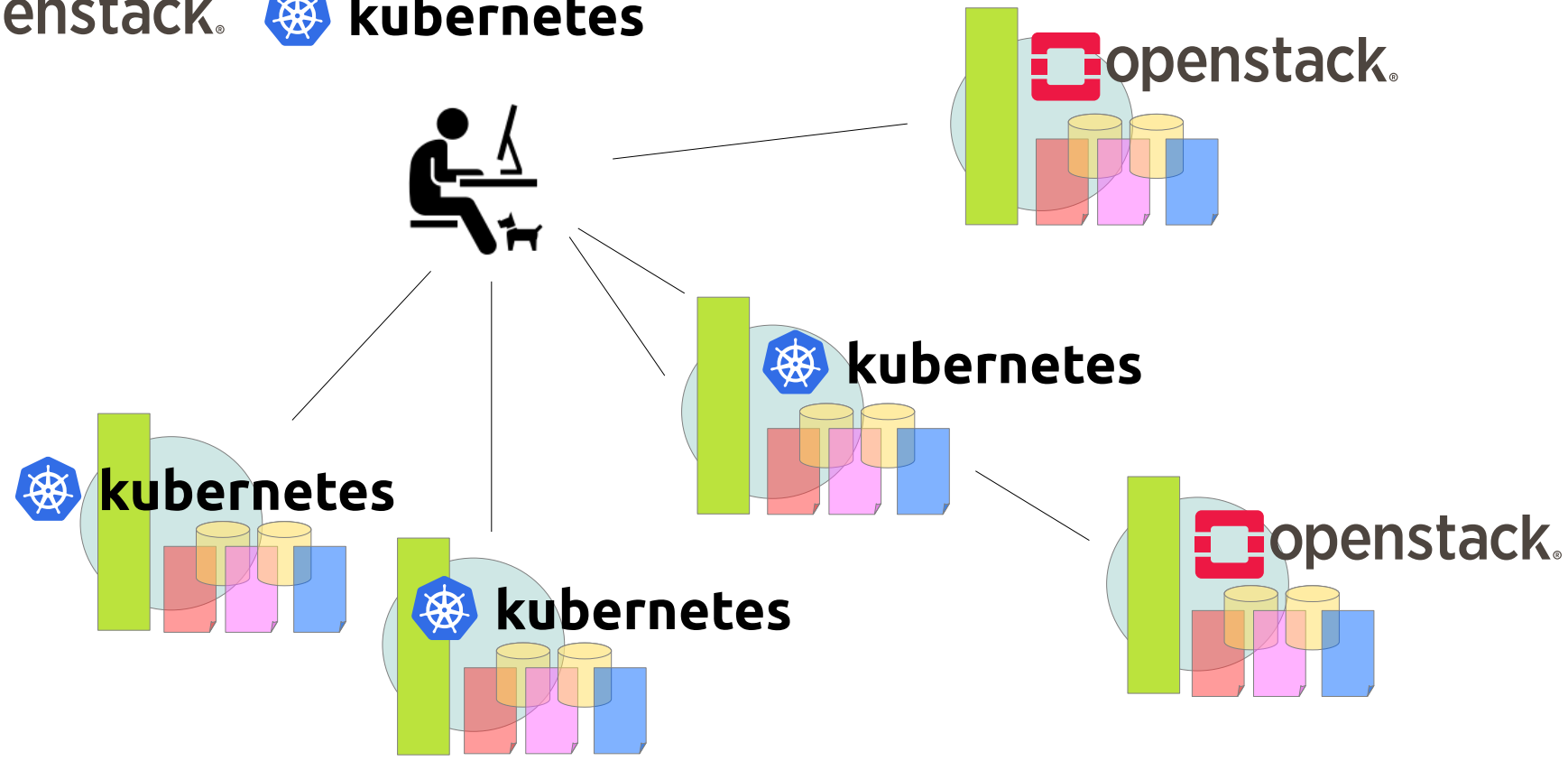
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Implementation

Cloud platforms January - March

Pick one as priority, one as stretch

openstack kubernetes



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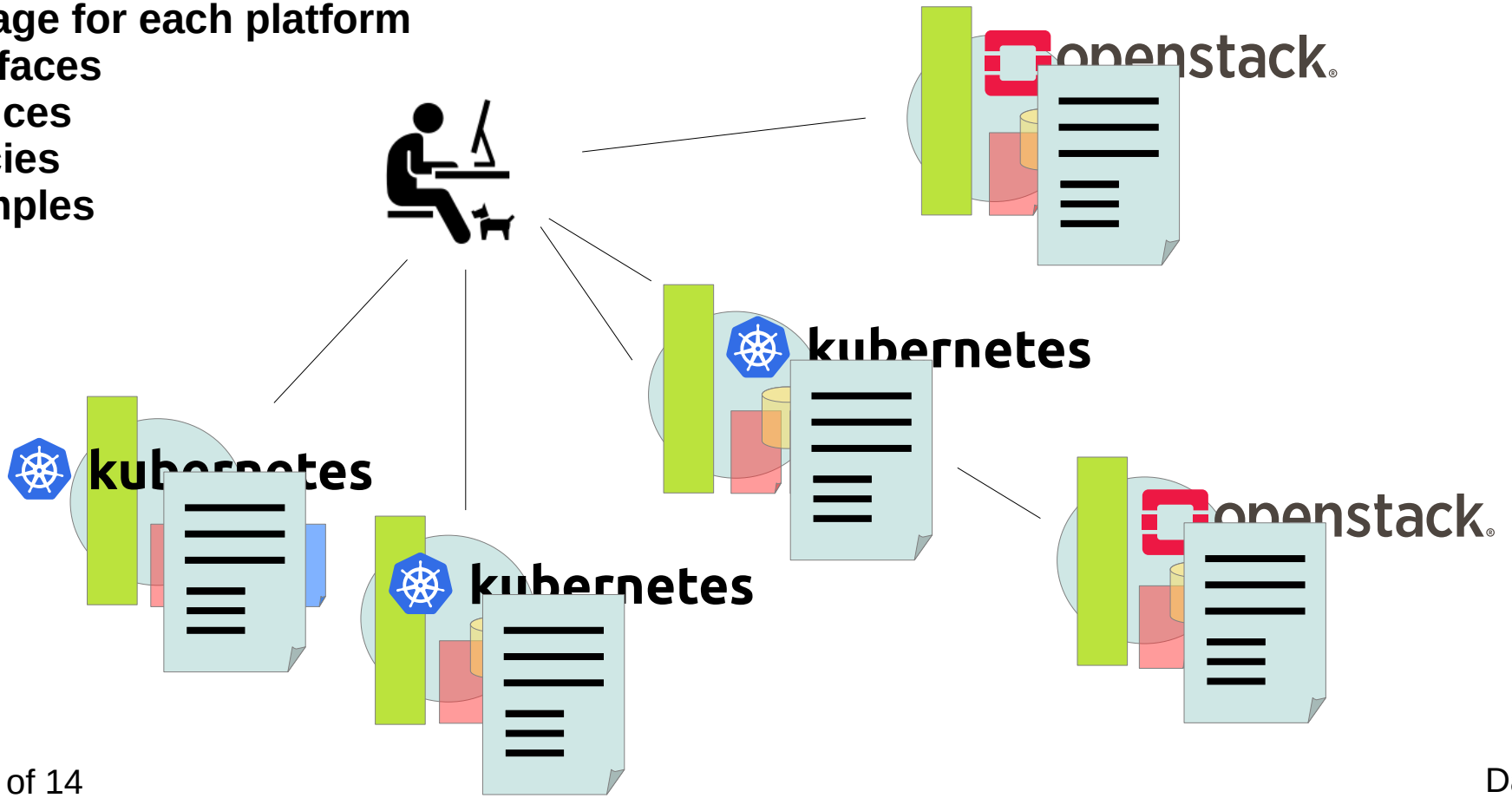
Help needed

Cloud platforms January - March

Pick one as priority, one as stretch

Wiki page for each platform

- **interfaces**
- **services**
- **policies**
- **examples**



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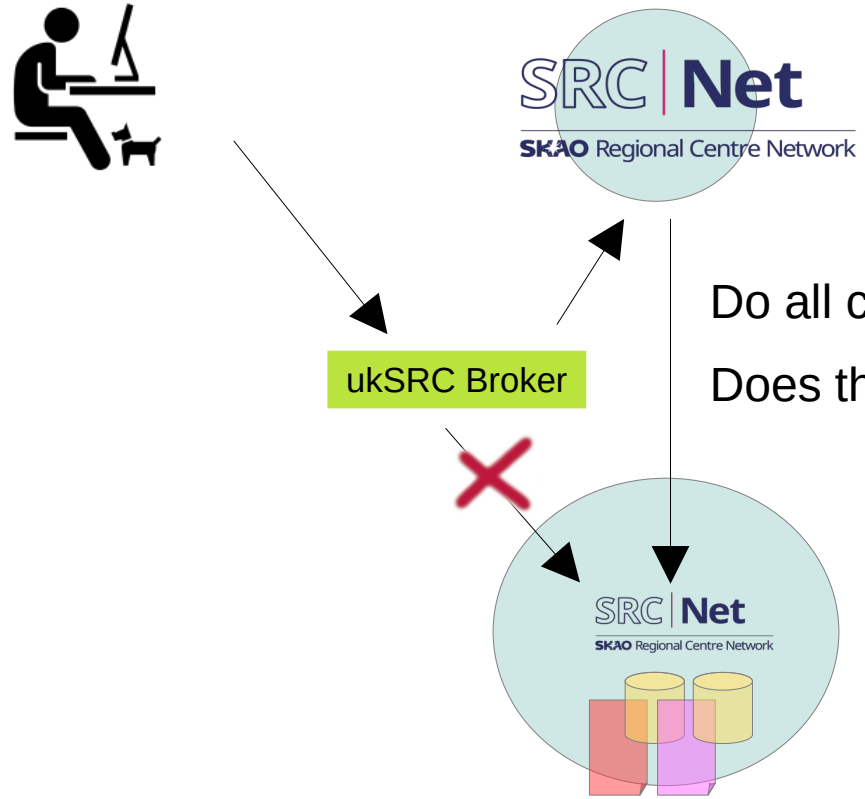
A solution - homogeneous platforms

Help needed

Integration with SRCNet

SRCNet - Centralized command and control

ukSRC - Distributed control, local policies



Do all calls have go go via the central service ?

Does this require two different tokens ?

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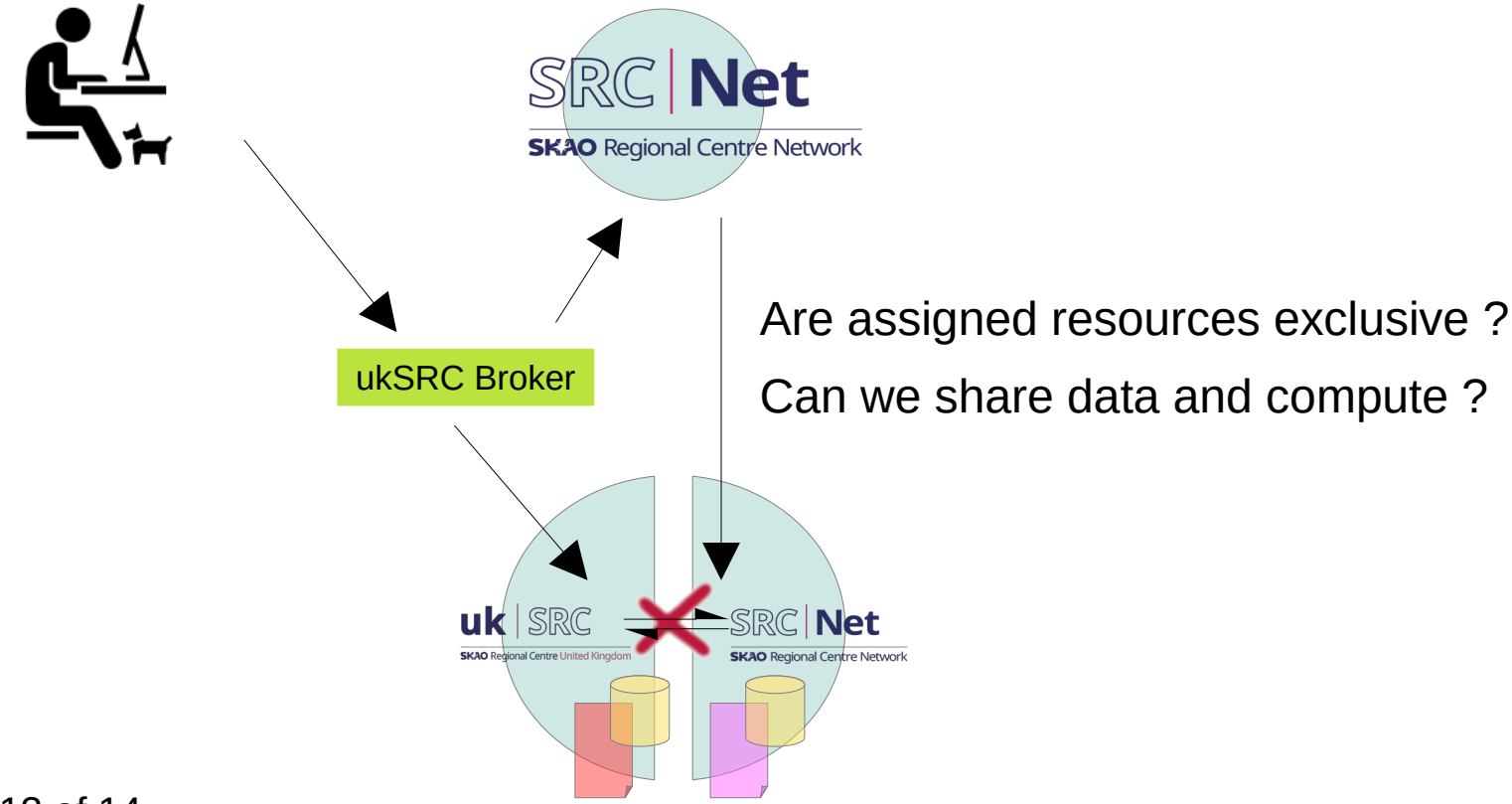
A solution - homogeneous platforms

Help needed

Integration with SRCNet

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Are assigned resources exclusive ?

Can we share data and compute ?

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Thank you

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GitHub 
https://github.com/uksrc/CalycopisBroker

Please join the discussion.

We need your help to make it happen.

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